(Cross-)Browser Fingerprinting via OS and Hardware Level Features

(Cross-)Browser Fingerprinting via OS and Hardware Level Features - Author not stated, NDSS Symposium.

- Published: date not stated
- Original: <https://www.ndss-symposium.org/ndss2017/ndss-2017-programme/cross-browser-fingerprinting-os-and-hardware-level-features/>
- Preserved from: https://www.ndss-symposium.org/ndss2017/ndss-2017-programme/cross-browser-fingerprinting-os-and-hardware-level-features/ (live) on 2026-08-08

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Download: *Paper (PDF)

**Date: **27 Feb 2017

**Document Type: **Reports

Additional Documents: *Slides Video

Associated Event: *NDSS Symposium 2017

Abstract:

In this paper, we propose a browser fingerprinting technique that can track users not only within a single browser but also across different browsers on the same machine. Specifically, our approach utilizes many novel OS and hardware level features, such as those from graphics cards, CPU, and installed writing scripts. We extract these features by asking browsers to perform tasks that rely on corresponding OS and hardware functionalities.

Our evaluation shows that our approach can successfully identify 99.24% of users as opposed to 90.84% for state of the art on single-browser fingerprinting against the same dataset. Further, our approach can achieve higher uniqueness rate than the only cross-browser approach in the literature with similar stability.